

Sweta Sahu

716-446-6484 | sahuswetaa23@gmail.com | [linkedin.com/in/23-swetasahu](https://www.linkedin.com/in/23-swetasahu) | github.com/sweta-sahu | [Portfolio](#)

New York, NY | Open to Relocation

EXPERIENCE

AI Software Engineer

Feb 2026 – Jul 2026

The Research Foundation for SUNY

Buffalo, NY

- Architected a multi-agent orchestration layer in **LangGraph** for an early-stage materials-science startup platform, combining deterministic routing with an LLM-driven planning loop to autonomously coordinate 15+ chained tool-calling steps across research workflows, reducing multi-step query latency by an estimated 40%.
- Designed and built a **vector database** pipeline via **FastAPI** to chunk, embed, and retrieve technical documents, integrating live tool-based **RAG** from JARVIS-DFT, Materials Project, and OPTIMADE across 1,000+ documents, improving retrieval accuracy by an estimated 35% against an internal evaluation set.
- Evaluated and deployed self-hosted ML models (**ALIGNNN**, **MACE**) for real-time property prediction, monitoring endpoint uptime and latency via **Grafana** – a deliberate architectural tradeoff over third-party model APIs to preserve data privacy, control cost, and avoid rate limits during high-volume research queries.
- Built a deterministic confidence-flagging framework (4 validation functions) to automatically detect cross-database conflicts, coverage gaps, and methodology mismatches, cutting flagged data-inconsistency issues by an estimated 50%.

Senior Research Aide

Feb 2025 – Dec 2025

The Research Foundation for SUNY

Buffalo, NY

- Architected a fault-tolerant, high-throughput distributed **IoT** pipeline via **AWS IoT Core**, **Kinesis**, and **Lambda** to process 1M+ telemetry events/day using MQTT/Modbus protocols, maintaining 99.9% uptime for mission-critical sensor data.
- Developed secure full-stack architectures using **React/Redux**, **Node.js (Express)**, and Python, integrating **AWS Cognito** for RBAC to support high-concurrency data streaming and enterprise-grade authentication.
- Engineered low-latency **WebSocket** pipelines for real-time analytics, supporting 500+ concurrent users at peak with sub-second state updates, reducing UI latency 30% under heavy production loads.
- Automated multi-region cloud infrastructure using **Terraform**, **Docker**, and **Kubernetes (EKS)**, reducing provisioning time 60%; enforced 95% unit/integration test coverage and automated **OWASP** security audits via **GitHub Actions CI/CD**, ensuring secure, zero-regression releases for containerized microservices.
- Increased system observability via **Grafana/CloudWatch** dashboards, reducing **MTTR** 25% through proactive anomaly detection; authored 15+ Technical Design Documents to accelerate developer onboarding.

Software Developer

June 2021 – July 2024

Tata Consultancy Services

Mumbai, India

- Owned backend architecture and production delivery for 3 high-traffic **Spring Boot** microservices, designing 25+ **REST APIs** and improving overall production reliability by 30% across the platform.
- Reduced inter-service latency an estimated 40% via an event-driven **Apache Kafka** system; diagnosed a consumer-lag incident from single-consumer topic growth, resolving it with a partitioned, parallel consumer group and a weekly retention policy.
- Lowered database load 45% and response times 60% via **Redis** distributed caching; diagnosed and fixed a production cache-consistency bug from delayed invalidation after **MongoDB** writes, correcting the write-to-cache flow and adding a regression test to prevent future recurrence in production.
- Acted as Backend Owner and Mentor, leading 10+ code reviews, driving Agile sprint delivery, and managing on-call rotations for services productionized on **AWS EC2**; drove production quality through **TDD (JUnit, Mockito)** to achieve 95%+ unit and integration test coverage across 10+ production microservices.
- Contributed to a **GNN**-based recommendation system, training graph embeddings over 100K+ products and evaluating ranking performance using offline metrics to reduce overfitting and improve generalization.

PROJECTS

WanderGenie – Agentic AI Travel Assistant | *LangGraph, FastAPI, React/TypeScript*

Nov 2025

- Served as LLM/Agent Lead on a 4-person team, designing a multi-agent trip-planning system (Planner, Researcher, Packager-Executor) with a hybrid **pgvector** + **Neo4j** retrieval layer and a full-stack **FastAPI/React** integration delivering real-time, context-aware itinerary edits and recommendations.

Incident Zero – Multimodal AI Security Investigator | *MCP, Python, Static Analysis*

Feb 2026 – Mar 2026

- Built a multimodal AI security-investigation platform coordinating 5 specialized **MCP** servers (CodeScan, LogReasoner, ScreenshotAnalyzer, DiagramExtractor, Patcher) to automate vulnerability detection and evidence correlation, auto-generating unified-diff security patches for identified vulnerabilities.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL

AI & Agentic Systems: LangGraph, LangChain, RAG, Vector DBs (pgvector, Neo4j), PyTorch, MCP

Backend & Frontend: FastAPI, Node.js (Express), Spring Boot, REST APIs, React, WebSockets, Kafka

Cloud, Data & DevOps: AWS (EC2, Lambda, S3, SageMaker), Docker, Kubernetes, Terraform, CI/CD

Databases & Testing: PostgreSQL, MongoDB, Redis, JUnit/Mockito, TDD, OWASP, OAuth2/OIDC

EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

M.S., Computer Science and Engineering (AI/ML Specialization) – GPA: 3.867/4.0

Aug 2024 – Dec 2025